METRIC
MIL-R-24727(NAVY)
31 August 1989

MILITARY SPECIFICATION

ROTARY JOINTS, FIBER OPTIC, SHIPBOARD (METRIC), GENERAL SPECIFICATION FOR

This specification is approved for use by the Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements of passive fiber optic rotary joints. The rotary joint is used to couple optical signals through a rotating device. The rotary joint is a passive device which allows transmission of signals over single or multiple fibers between rotating and stationary components. Excluded from this specification are rotary joints that contain a separate source and detector.

1.2 Classification. Rotary joints covered by this specification are classified as specified in 1.2.1 through 1.2.3.

1.2.1 Type. The type designation defines the number of optical fiber paths that are transmitted across the rotating interface.

Type S - Single fiber
Type M - Multiple fibers

1.2.2 Style. The style designation defines whether or not a fiber path occupies the axis of the rotating interface.

Style A - On-axis (see 6.6.4)
Style B - Off-axis (see 6.6.3)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.2.3 Port construction. The port construction designation defines the method of interfacing rotary joints to the fiber optic cables.

Port construction A - Cable pigtail, single-mode
Port construction B - Cable pigtail, multimode
Port construction C - Connectorized, single-mode
Port construction D - Connectorized, multimode

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-P-35	- Passivation Treatments for Corrosion-Resistant Steel.
TT-I-735	- Isopropyl Alcohol.

MILITARY

MIL-S-901	- Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for.
MIL-H-5606	- Hydraulic Fluid, Petroleum Base; Aircraft, Missile, and Ordnance.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4, JP-5 and JP-5/JP-8 ST.
MIL-F-16884	- Fuel, Naval Distillate.
MIL-L-17331	- Lubricating Oil, Steam Turbine and Gear, Moderate Service.
MIL-L-23699	- Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-R-24727/1	- Rotary Joints, Fiber Optic, Shipboard, Single Fiber, On-Axis, Multimode Cable Pigtail (Metric).
MIL-R-24727/2	- Rotary Joints, Fiber Optic, Shipboard, Multiple Fibers, On-Axis, Multimode Cable Pigtail (Metric).
MIL-C-55330	- Connectors, Electrical and Fiber Optic, Packaging of.
MIL-C-85045	- Cable, Fiber Optic, Shipboard (Metric) General Specification for.
MIL-C-85045/14	- Cable, Fiber Optic, Fiber Optic Cable Configuration Type B (Pigtail), Optical Fiber Type MM (Graded Index, Glass Core and Glass Cladding, Multimode). (Metric)

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STANDARDS

FEDERAL

FED-STD-313 - Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities.

MILITARY

MIL-STD-454 - Standard General Requirements for Electronic Equipment.
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines.
MIL-STD-883 - Test Methods and Procedures for Microelectronics.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.
MIL-STD-1344 - Test Methods for Electrical Connectors.
DOD-STD-1678 - Fiber Optics Test Methods and Instrumentation.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

455 - Standard Test Procedures for Fiber Optic Fibers, Cables, Transducers, Connecting and Terminating Devices. (DOD adopted)
455-2 - Impact Test Measurements for Fiber Optic Devices. (DOD adopted)
455-4 - Fiber Optic Connector/Component Temperature Life. (DOD adopted)
455-5 - Humidity Test Procedure for Fiber Optic Connecting Devices. (DOD adopted)
455-6 - Cable Retention Test Procedure for Fiber Optic Cable Interconnecting Devices. (DOD adopted)
455-12 - Fluid Immersion Test Procedure for Fiber Optic Connecting Devices.
455-13 - Visual and Mechanical Inspection of Fibers, Cables, Connectors and/or Other Fiber Optic Devices. (DOD adopted)
455-16 - Salt Spray (Corrosion) Test for Fiber Optic Components.
455-20 - Measurement of Change in Optical Transmittance. (DOD adopted)
455-22 - Ambient Light Susceptibility of Fiber Optic Components. (DOD adopted)

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EIA (Continued)

455-34 - Interconnection Device Insertion Loss Test. (DOD adopted)
455-35 - Fiber Optic Connector Dust (Fine Sand) Test.
455-36 - Twist Test for Fiber Optic Connecting Devices. (DOD adopted).
455-42 - Optical Crosstalk in Fiber Optic Components.
455-49 - Procedure to Measure Nuclear Radiation Effects in Fiber Optic Components.
455-71 - Procedure to Measure Temperature-Shock Effect on Fiber Optic Components.

(Application for copies should be addressed to the Electronic Industries Association, 2001 Eye Street, NW, Washington DC 20006.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. *REQUIREMENTS*

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.3 Materials. The rotary joint shall be constructed of material as specified (see 3.1). The materials selected shall have an intended lifetime of 20 years. They shall be of the lightest practical material suitable for its intended purpose. All combinations of materials shall be non-magnetic and non-nutrient. The materials shall have no adverse effect on the health of the personnel when used for its intended purpose. All specified materials are subject to toxicological data and formulations review and/or inspection for safety of the material by the Government.

3.3.1 Recovered materials. Unless otherwise specified herein, all material incorporated in the products covered by this specification shall be new. Products may be fabricated using raw materials produced from recovered bulk materials to the extent practicable if the intended use of the product is not jeopardized. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become part of a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of partially processed, assembled, used or rebuilt products are allowed under this specification.

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3.3.2 Metals. Unless otherwise specified (see 3.1), all exposed steel parts shall be corrosion-resistant and shall have a passivated surface that is compatible with external coatings or platings of the type and color specified.

3.3.3 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless finished to prevent electrolytic corrosion. Selection and protection of dissimilar metal combinations shall be in accordance with MIL-STD-889.

3.3.4 Finish. The resultant finish on all rotary joints shall be:

(a) Aluminum components - cadmium plate over electroless nickel for external parts.
(b) Stainless steel components - passivate in accordance with QQ-P-35.

3.3.5 Nonmetallic materials. Nonmetallic materials used in the construction of rotary joints shall not be degraded under the specified environmental conditions.

3.3.6 Material safety data sheet (MSDS). The contracting activity shall be provided a material safety data sheet at the time of contract award. The MSDS shall be provided in accordance with the requirements of FED-STD-313. The MSDS shall be included with each shipment of the material covered by this specification (see 6.7).

3.4 Design and construction. The configuration and physical dimensions of the rotary joints shall be as specified (see 3.1). The rotary joint shall consist of an assembly which allows transmission across a rotating interface. The rotary joint shall have one or more fiber paths with at least one input port and at least one output port. The external port construction shall consist of either connectors or pigtails as the interface to the fiber optic system as specified (see 3.1). Rotary joints with the same part number shall be physically and functionally interchangeable.

3.4.1 Fiber optic pigtail. Rotary joints with port construction A or B shall be supplied with two or more cable pigtails as specified (see 3.1). One end of the pigtail shall be permanently mounted in the rotary joint housing, and the mounting construction shall provide strain relief for the pigtail. The pigtail length shall be not less than 1 meter. Metal components shall not be used within the pigtail, except for mounting to the rotary joint housing or termination connector.

3.4.2 Fiber optic connectors. Rotary joints with port construction C or D shall be supplied with fiber optic connectors as specified (see 3.1).

3.4.3 Mounting hardware. The rotary joint shall be provided with mounting hardware as specified (see 3.1).

3.4.4 Seals. Grommets, O-rings, interface seals, boots, gaskets, or other sealing devices, as needed by the particular rotary joint design, shall be provided as specified (see 3.1).

3.4.5 Enclosures. The type of enclosure shall be either hermetic or watertight as specified (see 3.1) (see 3.5.3.5 and 3.5.3.6).

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3.4.6 Finish (see 4.6.1.1). The rotary joint finish shall have no peeling or chipping of the plating or finishes, nicks, burrs, or other surface blemishes.

3.4.7 Size (see 4.6.1.1). The size of the rotary joint shall be as specified (see 3.1).

3.4.8 Weight (see 4.6.2.1). The weight of the rotary joint shall be as specified (see 3.1).

3.4.9 Color (see 4.6.1.1). The color of the rotary joint shall be as specified (see 3.1).

3.4.10 Maintainability. The rotary joint shall require no preventive maintenance.

3.4.11 Bearings. If bearings are used in the design of the rotary joint, then those bearings shall be in accordance with MIL-STD-454, requirement 6.

3.5 Performance requirements. The performance requirements shall be defined in terms of optical, mechanical, environmental, and chemical properties.

3.5.1 Optical properties.

3.5.1.1 Insertion loss (see 4.6.3.1). The insertion loss (see 6.6.3) between each input port and each possible associated output port at any rate which is at or below the rated rotational rate shall be not greater than the limit specified (see 3.1).

3.5.1.2 Change in optical transmittance (see 4.6.3.2). The change in optical transmittance shall be not greater than 0.5 decibel (dB) for any environmental or mechanical requirement.

3.5.1.3 Return loss (see 4.6.3.3). The return loss (see 6.6.6) shall be not greater than minus 26 dB for rotary joints with single-mode fiber.

3.5.1.4 Ambient light susceptibility (see 4.6.3.4 and 6.6.1). The optical power from any one port or sum of ports shall be not greater than minus 50 decibels referred to 1 milliwatt (dBmW).

3.5.1.5 Crosstalk (see 4.6.3.5 and 6.6.2). The sum of the signal power levels for rotary joints with two or more signal paths shall be below the level of the active channel by at least 40 dB.

3.5.2 Mechanical properties.

3.5.2.1 Impact (see 4.6.4.1). The rotary joint shall not be visibly damaged or otherwise rendered unfit for operational use. The rotary joint shall maintain the minimum required rotational torque (see 3.5.2.5). The rotary joint shall meet the optical requirements of 3.5.1.2.

3.5.2.2 Twist (see 4.6.4.2). Cable-to-rotary joint seals shall not be rendered inoperable nor shall any other physical damage occur. The rotary joint shall meet the optical requirements of 3.5.1.2.

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3.5.2.3 Cable seal flexing (see 4.6.4.3). Cable-to-rotary joint strain relief mechanisms shall prevent loss of environmental sealing or other damage that may impair the rotary joint operation.

3.5.2.4 Crush resistance (see 4.6.4.4). The rotary joint shall show no evidence of broken parts or damage to seals. The rotary joint shall meet the optical requirements of 3.5.1.2.

3.5.2.5 Cable pull-out force (for port construction A and B only) (see 4.6.4.5). The minimum cable-to-rotary joint pull-out strength shall be 50 percent of the specified tensile strength of the attached cable or 73.5 kilograms (kg), whichever is less. There shall be no evidence of cable jacket damage, cable clamp failure, cable-to-rotary joint seal damage, distortion or bending of rotary joint parts, or cable disengagement from the clamp. The rotary joint shall meet the optical requirements specified in 3.5.1.2.

3.5.2.6 Connector pull-out force (for port construction C and D only) (see 4.6.4.6). The minimum connector-to-rotary joint pull-out strength shall be one-half the specified tensile strength of the attached connector or 73.5 kg, whichever is less. There shall be no evidence of connector-to-rotary joint seal damage, distortion or bending of rotary joint parts, or connector disengagement from the rotary joint. The rotary joint shall meet the optical requirements specified in 3.5.1.2.

3.5.2.7 Rotational variation and rated rotation rate (see 4.6.4.7). The rated rotational rate shall be as specified (see 3.1). For type S rotary joints, the insertion loss shall not change by more than 0.5 dB. For type M rotary joints, the insertion loss shall not change by more than 1.0 dB.

3.5.2.8 Rotational torque (see 4.6.4.8). The torque required to cause the rotary joint to rotate shall be as specified (see 3.1).

3.5.2.9 Mechanical life (see 4.6.4.9). The rotary joint shall meet the optical requirements of 3.5.1.2 for 5,000 hours of operation.

3.5.3 Environmental properties. The rotary joint shall meet all requirements specified (see 3.1) during the specified operating environments and after the specified storage environment. The operating temperature range and storage temperature range shall be as shown in table I, as specified (see 3.1).

TABLE I. Temperature ranges.

Range	Operating (°C)	Storage (°C)
1	-54 to +65	-62 to +71
2	-28 to +65	-62 to +71

3.5.3.1 Temperature life (see 4.6.5.1). The rotary joint shall show no evidence of dimensional change, opening of seals, cracking, or any form of physical damage which will impair performance. The rotary joint shall meet the optical requirements of 3.5.1.2. When a hermetic enclosure is specified, the rotary joint shall meet the requirements specified in 3.5.3.6.

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3.5.3.2 Thermal shock (see 4.6.5.2). The rotary joint shall be fully functional and show no evidence of physical degradation. The rotary joint shall meet the optical requirements of 3.5.1.2. When a hermetic enclosure is specified, the rotary joint shall meet the requirements specified in 3.5.3.6.

3.5.3.3 Temperature-humidity cycling (see 4.6.5.3). The rotary joint shall reveal no evidence of deterioration of parts or materials, loosening of finishes, physical distortion, corrosion of metals, entrapment of moisture, separation of bonded surfaces, or other physical damage. The rotary joint shall meet the optical requirements of 3.5.1.2.

3.5.3.4 Salt spray (corrosion) (see 4.6.5.4). The rotary joint shall show no evidence of salt penetration into the rotary joint seals nor peeling, flaking or color change in material as evidence of corrosive effects.

3.5.3.5 Water pressure (for watertight enclosures only) (see 4.6.5.5). The rotary joint shall reveal no evidence of the penetration of indicator dye into the sealed region of the rotary joint. The rotary joint shall meet the optical requirements of 3.5.1.2.

3.5.3.6 Hermetic (for hermetic enclosures only) (see 4.6.5.6). When a hermetic enclosure is specified, the rotary joint shall be permanently sealed to prevent transmission of air, gas, or moisture.

3.5.3.7 Freezing water (see 4.6.5.7). The rotary joint shall not be damaged upon exposure to freezing water. The rotary joint shall meet the optical requirements of 3.5.1.2.

3.5.3.8 Fluid immersion (see 4.6.5.8). The rotary joint shall reveal no swelling or softening of material, no loss of sealing capability or other effects detrimental to the operation of the rotary joint.

3.5.3.9 Sand and dust (see 4.6.5.9). The rotary joint shall continue to operate and show no damage as a result of sand and dust. The rotary joint shall meet the optical requirements of 3.5.1.2.

3.5.3.10 Shock (see 4.6.5.10). The rotary joint shall show no evidence of physical damage that impairs its specified function. The rotary joint shall meet the optical requirements of 3.5.1.2. When a hermetic enclosure is specified, the rotary joint shall meet the requirements specified in 3.5.3.6.

3.5.3.11 Vibration (see 4.6.5.11). The rotary joint shall show no evidence of physical damage that impairs its specified function. The rotary joint shall meet the optical requirements of 3.5.1.2. When a hermetic enclosure is specified, the rotary joint shall meet the requirements specified in 3.5.3.6.

3.5.3.12 Nuclear radiation resistance (see 4.6.5.12). Unless otherwise specified (see 3.1), the rotary joint shall meet the requirements of 3.5.1.2 when tested in accordance with 4.6.5.12.

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3.5.4 Chemical properties.

3.5.4.1 Fungus resistance (see 4.6.6.1). The rotary joint shall be in accordance with MIL-STD-454, requirement 4, for fungus-inert materials or grade 1 classification of MIL-STD-810, method 508.

3.5.4.2 Flame survivability (see 4.6.6.2). The splice shall meet the burning resistance criteria of condition B for flame extinguishing time after removal of applied flame, and there shall be no violent burning or explosive type fire. The splice shall meet the optical requirements of 3.5.1.2.

3.6 Safety. The handling, installation, and operation of the rotary joint shall not endanger personnel by direct contact or by long-term effects due to deterioration of any component within the rotary joint construction. The rotary joint design and construction shall conform to the safety standards specified in MIL-STD-454, requirement 1.

3.7 Transportability. The rotary joint operational performance specified herein shall not be affected following prolonged storage or shipment under environmental conditions for shipment and storage specified herein, and as required for packaging for shipment (see 5.1).

3.8 Identification marking (see 4.6.1.1). Unless otherwise specified (see 3.1), the rotary joint shall be identified by a marking applied to the outer surface of the rotary joint. The identification marking shall consist of the following:

(a) Specification sheet part number.
(b) Manufacturer's code designation.
(c) The words "Fiber Optic Rotary Joint".
(d) Date code (4 digit - week, year).
(e) Port identification (input and output ports).

The characteristics and method of identification marking shall conform to requirements of MIL-STD-1285. The identification marking shall be permanent and legible in accordance with the marking requirements of MIL-STD-1285. Identification marking shall be applied with the vertical axes of the printed characters perpendicular to the longest dimension of the rotary joint housing. All characters shall remain legible for the intended lifetime of the rotary joint. Except as otherwise specified (see 3.1), the rotary joint accessory parts shall also be identified by a legible and permanent marking applied in accordance with MIL-STD-1285.

3.9 Workmanship (see 4.6.1.1). The rotary joint shall be free of manufacturing flaws that would degrade performance, inhibit proper connection to interfacing elements, and otherwise yield an inferior product. The following are minimal considerations and are not intended to exclude other pertinent workmanship requirements deemed necessary by the contractor:

(a) Outer surface shall be free of rust, burnt areas, abrasions, holes, roughhead areas, bulges, thin spots, and indentations.
(b) Inner parts shall be free of fractures, holes, discontinuities, bulges, and thin spots.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.4).
(b) Quality conformance inspection (see 4.5).

4.2.1 Toxicological data and formulation. The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in applicable portions of EIA 455.

4.4 First article inspection. First article inspection shall consist of all the tests listed in table II and as specified (see 3.2).

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TABLE II. First article inspection.

Inspection	Requirement	Test method	Sample size
Group I			
Visual and mechanical	3.4, 3.9	4.6.1.1	9
Finish	3.4.6	4.6.1.1	1/
Size	3.4.7	4.6.1.1	1/
Weight	3.4.8	4.6.2.1	1/
Color	3.4.9	4.6.1.1	1/
Identification marking	3.8	4.6.1.1	1/
Insertion loss	3.5.1.1	4.6.3.1	1/
Return loss	3.5.1.3	4.6.3.3	1/
Ambient light susceptibility	3.5.1.4	4.6.3.4	1/
Crosstalk	3.5.1.5	4.6.3.6	1/
Group II			
Water pressure	3.5.3.5	4.6.5.5	1
Hermetic	3.5.3.6	4.6.5.6	1
Flame survivability	3.5.4.2	4.6.6.2	1
Group III			
Temperature life	3.5.3.1	4.6.5.1	1
Thermal shock	3.5.3.2	4.6.5.2	2/
Temperature-humidity cycling	3.5.3.3	4.6.5.3	2/
Salt spray	3.5.3.4	4.6.5.4	2/
Freezing water	3.5.3.7	4.6.5.7	2/
Fluid immersion	3.5.3.8	4.6.5.8	2/

See footnotes at end of table.

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TABLE II. First article inspection - Continued.

Inspection	Requirement	Test method	Sample size
Sand and dust	3.5.3.9	4.6.5.9	2/
Shock	3.5.3.10	4.6.5.10	1
Vibration	3.5.3.11	4.6.5.11	3/
Nuclear radiation resistance	3.5.3.12	4.6.5.12	3
Fungus resistance	3.5.4.1	4.6.6.1	4/
Group IV			
Impact	3.5.2.1	4.6.4.1	1
Twist	3.5.2.2	4.6.4.2	5/
Cable seal flexing	3.5.2.3	4.6.4.3	5/
Crush resistance	3.5.2.4	4.6.4.4	5/
Cable pull-out force	3.5.2.5	4.6.4.5	5/
Connector pull-out force	3.5.2.6	4.6.4.6	5/
Rotational variation and rated rotation rate	3.5.2.7	4.6.4.7	5/
Rotational torque	3.5.2.8	4.6.4.8	5/
Mechanical life	3.5.2.9	4.6.4.9	5/

1/ The same samples shall be used as in the visual and mechanical inspection.
2/ The same sample shall be used as in the temperature life inspection.
3/ The same sample shall be used as in the shock inspection.
4/ One sample shall be used from the nuclear radiation inspection.
5/ The same sample shall be used as in the impact inspection.

4.4.1 Sample. A rotary joint sample shall be submitted for each rotary joint construction for which first article approval is desired. The sample submitted shall be nine rotary joints of the same part number.

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4.4.2 Inspection routine. The sample shall be subjected to the inspections specified in table II in the order shown. All sample units shall be subjected to the inspection of group I. The sample shall then be divided into three groups. The sample units shall then be subjected to the inspection for their particular group. After completion of sample testing, all units shall be resubjected to group I testing. Any rotary joint failing any inspection shall not be subjected to further inspection.

4.4.3 Failures. One or more failures shall be sufficient cause for refusal to grant first article approval.

4.5 Quality conformance inspection. Quality conformance inspection shall consist of the inspections and tests specified for group A inspection (table III), group B inspection (table IV), and group C inspection (table V) (see 6.2).

4.5.1 Group A inspection. Group A inspections shall follow the order shown in table III.

TABLE III. Group A inspection.

Inspection	Requirement	Test method
Visual and mechanical	3.4, 3.9	4.6.1.1
Finish	3.4.6	4.6.1.1
Size	3.4.7	4.6.1.1
Weight	3.4.8	4.6.2.1
Color	3.4.9	4.6.1.1
Identification marking	3.8	4.6.1.1
Insertion loss	3.5.1.1	4.6.3.1
Return loss	3.5.1.3	4.6.3.3
Ambient light susceptibility	3.5.1.4	4.6.3.4
Crosstalk	3.5.1.5	4.6.3.6
Rotational variation and rated rotation rate	3.5.2.7	4.6.4.7

4.5.2 Group B inspection. Group B inspection shall consist of the inspections specified in table IV. Group B inspections shall be made on units that have passed the group A inspection.

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TABLE IV. Group B inspection.

Inspection	Requirement	Test method
Shock	3.5.3.10	4.6.5.10
Thermal shock	3.5.3.2	4.6.5.2
Twist	3.5.2.2	4.6.4.2
Cable seal flexing	3.5.2.3	4.6.4.3
Cable pull-out force	3.5.2.5	4.6.4.5
Connector pull-out force	3.5.2.6	4.6.4.6

4.5.3 Group C inspections. Group C inspections shall consist of the *inspections in table V. Group C inspections shall have satisfactorily completed* all group A and group B inspections. After completion of group C inspections, the samples shall be resubjected to group A inspection.

TABLE V. Group C inspection.

Inspection	Requirement	Test method
Water pressure	3.5.3.5	4.6.5.5
Hermetic	3.5.3.6	4.6.5.6
Flame survivability	3.5.4.2	4.6.6.2
Temperature life	3.5.3.1	4.6.5.1
Temperature-humidity cycling	3.5.3.3	4.6.5.3
Salt spray	3.5.3.4	4.6.5.4
Freezing water	3.5.3.7	4.6.5.7
Fluid immersion	3.5.3.8	4.6.5.8
Sand and dust	3.5.3.9	4.6.5.9
Nuclear radiation resistance	3.5.3.12	4.6.5.12

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TABLE V. Group C inspection - Continued.

Inspection	Requirement	Test method
Fungus resistance	3.5.4.1	4.6.6.1
Vibration	3.5.3.11	4.6.5.11
Impact	3.5.2.1	4.6.4.1
Crush resistance	3.5.2.4	4.6.4.4
Rotational torque	3.5.2.8	4.6.4.8
Mechanical life	3.5.2.9	4.6.4.9

4.5.3.1 Disposition of group C inspection sample units. Sample units subjected to group C inspection that have failed shall not be delivered.

4.6 Methods of inspection.

4.6.1 General testing practice.

4.6.1.1 Visual and mechanical inspections. Unless otherwise specified (see 3.1), visual and mechanical examinations shall be performed in accordance with EIA 455-13, to verify that the design, construction, physical characteristics, dimensions, marking, and workmanship are in accordance with the applicable requirements.

4.6.1.2 Optical test methods. The use of optical test methods to assess the influence of test exposures on performance of rotary joints requires that they be connected to lengths of fiber. Unless otherwise specified (see 3.1), the optical properties inspections shall be performed at a wavelength of 1.310 ± 0.020 micrometers (μm). The rotary joints should be attached to lengths of cable of the type specified in MIL-C-85045. The cable lengths shall be not greater than 30 meters. The optical fiber components exposed at the test set (source, detector, or coupler) may be connectorized for access to the test set. For multimode fiber, the source used shall be noncoherent. Light launch conditions shall be in accordance with table VI.

TABLE VI. Light launch conditions.

Fiber type	Launch condition
single-mode	30-mm diameter mandrel
multimode	70/100 restricted, or 70/70 restricted (see 6.6.8 and 6.6.7)

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Any optical power detection method may be used if the method is sufficiently sensitive to measure the differential optical power levels as specified (see 3.5.1.1), and if the method provides repeatable readings (less than 3 percent variation).

4.6.1.3 Equivalent test methods. The use of equivalent test methods is allowed when specified (see 3.1 and 6.2).

4.6.2 Design and construction inspection.

4.6.2.1 Weight (see 3.4.8). The rotary joint weight shall be tested using scales of sufficient range and accuracy appropriate for the specified weight and tolerance.

4.6.3 Optical properties inspection.

4.6.3.1 Insertion loss (see 3.5.1.1). The insertion loss shall be measured in accordance with EIA 455-34.

4.6.3.2 Change in optical transmittance (see 3.5.1.2). The change in optical transmittance shall be measured in accordance with EIA 455-20 or equivalent (see 4.6.1.3), using a monitor fiber.

4.6.3.3 Return loss (see 3.5.1.3). The return loss shall be measured in accordance with appendix A of this specification (see 6.3 and appendix B).

4.6.3.4 Ambient light susceptibility (see 3.5.1.4). The ambient light susceptibility shall be measured in accordance with EIA 455-22, with the exception that the output power in the "on" state shall be referenced to 1 milliwatt. The test temperature shall be 25 degrees Celsius (°C).

4.6.3.5 Crosstalk (see 3.5.1.5). The rotary joint crosstalk shall be measured in accordance with EIA 455-42. The test temperature shall be 25°C.

4.6.4 Mechanical properties inspection.

4.6.4.1 Impact (see 3.5.2.1). The rotary joint shall be tested in accordance with EIA 455-2, moderate service class. The change in optical transmittance shall be measured after the test (see 4.6.3.2). At the completion of the test, the rotary joint shall be visually examined in accordance with 4.6.1.1.

4.6.4.2 Twist (see 3.5.2.2). The rotary joint shall be tested in accordance with EIA 455-36, for 50 cycles, with the exception that the test sample shall be a properly pigtailed rotary joint. The tensile load shall be 5 newtons and the number of loads to be applied shall be one. The change in optical transmittance shall be measured during and after the test (see 4.6.3.2). At the completion of the test, the rotary joint shall be visually examined in accordance with 4.6.1.1.

4.6.4.3 Cable seal flexing (for pigtailed rotary joints only) (see 3.5.2.3). The rotary joint shall be tested in accordance with MIL-STD-1344, method 2017, with the exception that the test sample shall be a properly pigtailed rotary joint. At the completion of the test, the rotary joint shall be visually examined in accordance with 4.6.1.1.

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4.6.4.4 Crush resistance (see 3.5.2.4). The rotary joint shall be tested in accordance with MIL-STD-1344, method 2008, with the exception that the test sample shall be a properly pigtailed rotary joint. The test load shall be 1250 newtons, and the number of loads shall be one. The change in optical transmittance shall be measured during and after the test (see 4.6.3.2). At the completion of the test, the rotary joint shall be visually examined in accordance with 4.6.1.1.

4.6.4.5 Cable pull-out force (see 3.5.2.5). The rotary joint shall be tested in accordance with EIA 455-6. The axial tensile load shall be applied to the load specified and shall be maintained for 10 minutes. The change in optical transmittance shall be measured during and after the test (see 4.6.3.2). At the completion of the test, the rotary joint shall be visually examined in accordance with 4.6.1.1.

4.6.4.6 Connector pull-out force (see 3.5.2.6). The rotary joint shall be tested in accordance with EIA 455-6, with the exception that the test sample shall be a connector on the rotary joint. The axial tensile load shall be applied to the load specified and shall be maintained for 10 minutes. The change in optical transmittance shall be measured during and after the test (see 4.6.3.2). At the completion of the test, the rotary joint shall be visually examined in accordance with 4.6.1.1.

4.6.4.7 Rotational variation and rated rotation rate (see 3.5.2.7). The rotary joint shall be rotated at 10 percent of its rated rotational rate. The speed of rotation shall be increased 10 percent per minute until the rated rotational rate is reached. The change in optical transmittance shall be measured during and after the test (see 4.6.3.2).

4.6.4.8 Rotational torque (see 3.5.2.8). The rotary joint shall be rotated using a torque wrench and the force necessary to initiate rotation shall be measured.

4.6.4.9 Mechanical life (see 3.5.2.9). The rotary joint shall be subjected to 5,000 hours of continuous operation at the rotation rate as specified (see 3.1) at ambient temperature of 65°C.

4.6.5 Environmental properties inspection. Unless otherwise specified (see 3.1), the rotary joint shall be nonrotating for the environmental tests.

4.6.5.1 Temperature life (see 3.5.3.1). The rotary joint shall be tested in accordance with EIA 455-4. The exposure time shall be 96 hours, and the exposure temperature shall be 105°C. The rotary joint shall be nonrotating during this test. The change in optical transmittance shall be measured after the test (see 4.6.3.2). At the completion of the test, the rotary joint shall be visually examined in accordance with 4.6.1.1 and tested for hermeticity (see 4.6.5.6), if applicable to the rotary joint design.

4.6.5.2 Thermal shock (see 3.5.3.2). The rotary joint shall be tested for 10 cycles in accordance with EIA 455-71. The high and low temperatures shall correspond to the specified storage temperature extremes. The change in optical transmittance shall be measured after the test (see 4.6.3.2). At the completion of the test, the rotary joint shall be visually examined in accordance with 4.6.1.1 and tested for hermeticity (see 4.6.5.6), if applicable to the rotary joint design.

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4.6.5.3 Temperature-humidity cycling (see 3.5.3.3). The rotary joint shall be tested in accordance with EIA 455-5. The relative humidity shall be 95 percent, and the temperature range shall be the specified operating temperature range. The change in optical transmittance shall be measured during and after the test (see 4.6.3.2). At the completion of the test, the rotary joint shall be visually examined in accordance with 4.6.1.1.

4.6.5.4 Salt spray (corrosion) (see 3.5.3.4). The rotary joint shall be tested in accordance with EIA 455-16. The exposure time shall be 96 hours and the exposure temperature shall be 35°C. At the completion of the test, the rotary joint shall be visually examined in accordance with 4.6.1.1.

4.6.5.5 Water pressure (see 3.5.3.5). The rotary joint shall be immersed in an aqueous dye penetrant solution at an applied pressure of 0.1 megapascal (equivalent to a depth of 10.4 meters) for 48 hours at a temperature of 10 to 35°C. The change in optical transmittance shall be measured after the test (see 4.6.3.2). At the completion of the test, the rotary joint shall be visually examined in accordance with 4.6.1.1.

4.6.5.6 Hermetic (see 3.5.3.6). The rotary joint shall be tested for hermetic seal in accordance with MIL-STD-883, method 1014. Test condition A1 shall be used for the seal test, and test condition C shall be used for the gross leak test.

4.6.5.7 Freezing water (see 3.5.3.7). The rotary joint shall be tested in accordance with DOD-STD-1678, method 4050. The rotary joint shall be completely immersed in water for 6 hours at the specified minimum operating temperature. The vessel containing the water need not be sealed. The size (volume) of the vessel shall be approximately two to three times the size of the rotary joint under test. The rotary joint shall be nonrotating during this test. The change in optical transmittance shall be measured after the test (see 4.6.3.2). At the completion of the test, the rotary joint shall be visually examined in accordance with 4.6.1.1.

4.6.5.8 Fluid immersion (see 3.5.3.8). The rotary joint shall be tested in accordance with EIA 455-12. Each rotary joint shall be immersed in each immersion fluid as specified in table VII for a period of 24 hours. Each rotary joint shall be completely dried before further testing. At the completion of the test, the rotary joint shall be visually examined in accordance with 4.6.1.1.

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TABLE VII. Immersion test fluids.

Fluids	Specification	Test temperature (°C)
Fuel oil	MIL-F-16884	33 - 37
Turbine fuel (JP-5)	MIL-T-5624	20 - 25
Isopropyl alcohol	TT-I-735	20 - 25
Hydraulic fluids	MIL-H-5606	48 - 50
Lubricating oils	MIL-L-17331 MIL-L-23699	73 - 77
Coolant 1/	-	20 - 25
Sea water	-	20 - 25

1/ Monsanto Coolanol 25 or equivalent.

4.6.5.9 Sand and dust (see 3.5.3.9). The rotary joint shall be tested in accordance with EIA 455-35. The change in optical transmittance shall be measured during and after the test (see 4.6.3.2). At the completion of the test, the rotary joint shall be visually examined in accordance with 4.6.1.1.

4.6.5.10 Shock (see 3.5.3.10). The rotary joint shall be tested in accordance with MIL-S-901, grade A, type A, class I. Unless otherwise specified (see 3.1), the change in optical transmittance shall be monitored continuously during and after the test with equipment having a time resolution of at least 50 microseconds (μs) (see 4.6.3.2). At the completion of the test, the rotary joint shall be visually examined in accordance with 4.6.1.1 and tested for hermeticity (see 4.6.5.6), if applicable to the rotary joint design.

4.6.5.11 Vibration (see 3.5.3.11). The rotary joint shall be tested in accordance with MIL-STD-1344, method 2005, test conditions II and VI. For test condition VI, test condition C shall apply. Test condition I shall be used during the low frequency portion of test condition II. Unless otherwise specified (see 3.1), the change in optical transmittance shall be monitored continuously during and after the test with equipment having a time resolution of at least 50 μs (see 4.6.3.2). At the completion of the test, the rotary joint shall be visually examined in accordance with 4.6.1.1 and tested for hermeticity (see 4.6.5.6), if applicable to the rotary joint design.

4.6.5.12 Nuclear radiation resistance (see 3.5.3.12). The rotary joint shall be tested in accordance with EIA 455-49. Unless otherwise specified (see 3.1), tests shall be performed at a wavelength of 1.310 ± 0.020 μm and at the low operating temperature, 20°C, and the high operating temperature. The total dose shall be as specified (see 6.2). The change in optical transmittance shall be measured during and after the test (see 4.6.3.2).

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4.6.6 Chemical properties inspection.

4.6.6.1 Fungus resistance (see 3.5.4.1). Rotary joints that do not meet the requirements of fungus-inert materials in accordance with MIL-STD-454, requirement 4, shall be tested for exposure to fungus in accordance with MIL-STD-810, method 508.

4.6.6.2 Flame survivability (see 3.5.4.2). The rotary joint shall be tested in accordance with MIL-STD-1344, method 1012, test condition A, with the exception that the test sample shall be a rotary joint instead of a connector. The change in optical transmittance shall be measured after the test (see 4.6.3.2).

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.8.)

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-C-55330. The level of packaging shall be as specified (see 6.2).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The rotary joints covered by this specification are intended for use in any application where their performance characteristics are required. The rotary joints are for installation in shipboard systems within the limitations of applicable performance requirements.

6.1.1 Engineering information. For rotary joints used in outside applications subject to freezing, the torque required to break the ice load should be specified (see 4.6.5.7).

6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of the specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
(c) Applicable specification sheet number, title, and date.
(d) When first article is required (see 3.2).
(e) Quality Conformance Program Information (see 4.6).
(f) Equivalent test methods, if other than as specified (see 4.6.1.3).

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(g) Total dose radiation requirement (see 4.6.5.12).
(h) Level of packaging required (see 5.1).
(i) Applicable specification sheet part number (see 6.9).
(j) Quantity of rotary joints required.

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DoD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.6.3.3 and appendix B	DI-MISC-80678	Certification/data report	---

The above DID's were those cleared as of the date of this specification. The current issue of DoD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 First article. When first article inspection is required, the items should be a first article sample. The first article should consist of 8 units. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Quality conformance. Quality conformance requires contractual definition of the overall test program, including sample sizes and lot sizes, if appropriate (see 4.5).

6.6 Definitions.

6.6.1 Ambient light susceptibility. Ambient light susceptibility is the optical power (dBmW) that enters the rotary joint from ambient illumination incident upon the rotary joint.

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6.6.2 Crosstalk. A measure of the optical power picked up by an optical fiber or path from another fiber or optical path. If the optical fiber or path at the receiving end of the energized path is P_o and the power in the disturbed fiber at the corresponding end is P_d then:

$$\text{crosstalk} = 10 \log_{10} \frac{P_d}{P_o} \text{ in dB}$$

Since P_d is always less than P_o, the crosstalk is always negative, the negative sign commonly ignored.

6.6.3 Insertion loss. Insertion loss is the radiant power loss (dB) caused by absorption, scattering, diffusion, leaky waves, dispersion, microbends, macrobends, reflection, radiation, or other causes when a rotary joint is inserted into the system.

6.6.4 Off-axis rotary joint. An off-axis rotary joint is a rotary joint that does not have an optical fiber path that occupies the axis of the rotating interface.

6.6.5 On-axis rotary joint. An on-axis rotary joint is a rotary joint that has an optical fiber path that occupies the axis of the rotating interface.

6.6.6 Return loss. Return loss is the optical power (dB) that is reflected back toward the source of optical power by the rotary joint.

6.6.7 70/70 restricted launch. A 70/70 restricted launch is a beam optics launch with a 70 percent spot size and source aperture equal to 70 percent of the fiber numerical aperture.

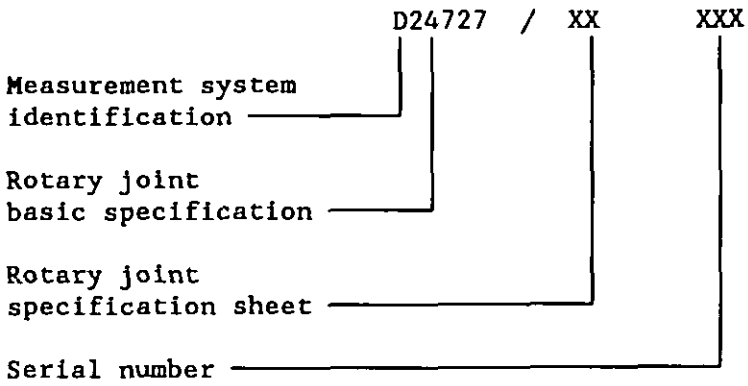
6.6.8 70/100 restricted launch. A 70/100 restricted launch is a beam optics launch with a 70 percent spot size and source aperture equal to or greater than the fiber numerical aperture.

6.7 Material Safety Data Sheets. Contracting officers will identify those activities requiring copies of completed Material Safety Data Sheets prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.8 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.9 Part or Identifying Number (PIN). Rotary joints should contain only the following:

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Examples:

D24727/01-001
D24727/02-002

6.10 Part designator. A part designator, if specified (see 3.1), should include classifications (see 1.2) as follows:

(a) Type (see 1.2.1).
(b) Style (see 1.2.2).
(c) Port construction (see 1.2.3).

Example: SAB.

6.11 Specification use. When MIL-C-85045 is referenced, the Navy version of this specification (Cable, Optical, General Specification for) should be used.

6.12 Subject term (key word) listing.

Change in optical transmittance
Fiber optic communications
Fiber optic component
Fiber optic rotating interfaces
Insertion loss
Mechanical life
Off-axis rotary joint
On-axis rotary joint
Optical slip ring
Rotational variation

Review activities:
Navy - EC, YD

Preparing activity:
Navy - SH
(Project 6004-N001)

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APPENDIX A

TEST PROCEDURE TO MEASURE RETURN LOSS

10. SCOPE

10.1 Scope. The intent of this procedure is to determine the ratio of optical power reflected by a component or an assembly to the optical power incident on a component port when that component assembly is introduced into a link or system. This ratio is termed return loss. This appendix is a mandatory part of this specification. The information contained herein is intended for compliance.

10.2 Test. The test for return loss is based on two setups. The first test setup shown on figure 1 is designed to measure the insertion loss of a coupler along a particular through a path of that device. The coupler is an integral part of measuring the insertion loss of the test setup shown on figure 2.

20. APPLICABLE DOCUMENTS

This section is not applicable to this specification.

30. TEST EQUIPMENT

30.1 Apparatus. The apparatus shall consist of the following primary items and those additional items identified on figures 1 and 2 and the specification sheet. The apparatus configuration shall be as shown on figures 1 and 2.

30.2 Optical power source. The optical power source shall consist of a light source emitting wavelengths suitable for the devices tested and can either be modulated or unmodulated. A light-emitting diode (LED) or injection laser with suitable radiation band is a suitable source and may be specified. The source shall be stable to plus or minus 1 percent (0.05 dB) over the measurement period. Wavelength selective filters may be used if necessary to limit optical spectral width.

30.3 Optical directional coupler. The directional coupler (or splitter) may be of any convenient coupling ratio. Its return path to forward path power ratio (directivity) when terminated in a nonreflecting termination shall be 10 dB greater than the maximum return loss ratio desired to be measured. The coupler shall be mode insensitive.

30.4 Test jumper assemblies. One pair of pigtails are required for each device under test (DUT) measurement.

30.5 Termination. An optical load which is fully absorbent and nonreflective (at least 10 dB below the desired level of test sensitivity) when coupled to the connecting cable of the test system. Index matching gel or fluid is a suitable termination.

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30.6 Optical power meter. A device capable of measuring incident optical power which is linear to within 5 percent over the power levels to be measured and stable to plus or minus 1 percent (0.05 dB) over the measurement period. Relative power levels only are required, so no absolute accuracy is necessary.

30.7 Interconnects/splices. Since all connectors used in the test will influence the test results, extremely low loss and repeatable units shall be used. Fusion splices are recommended if practical and are noted as such on the figures. Index matching materials at all interfaces not under test are highly recommended.

30.8 Mode filter. A 50-millimeter (mm) diameter loop of fiber/cable of the same type used in the coupler.

40. TEST SAMPLE

40.1 Test sample. The test sample or DUT shall be as specified (see 3.1).

50. TEST PROCEDURE

50.1 Determination of insertion loss of coupler. The test procedure shall be as follows:

(a) Referring to the test configuration shown in step A of figure 1, 1 meter of single-mode fiber is fusion-welded to the lead length of the fiber from port 3 of the coupler as a temporary joint. This fiber lead from port 2 is connected to a detector/power meter. Index matching material is applied to fiber leads from ports 1 and 4 to provide termination points. A power measurement is taken at the specified wavelength and a measurement, P_2, is recorded.

(b) In step B, cut the fiber between the coupler and the fusion joint, just beyond the joint, allowing material for polishing of the cut end. The free end of the fiber shall be properly prepared so that it is smooth and perpendicular to the axis. Connect the prepared end of the fiber to the detector ensuring that the power level is repeatable. Measure and record this power measurement as P_3. The insertion loss of this particular path between port 2 and port 3 of the coupler is given by the formula:

$$L_c = -10\log_{10} (P_2/P_3)$$

50.2 Determination of DUT return loss. The test procedure shall be as follows:

(a) Using the setup shown on figure 2, connect the fiber lead of port 1 of the coupler to the light source. Connect the detector/power meter to the fiber lead from port 3. Terminate the fiber leads from port 2 and 4 with index matching gel. Measure and record the insertion loss between ports 1 and 3 as P_o.

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(b) Switch the detector/power meter to the fiber lead from port 2 of the coupler and terminate the fiber from port 3 with index gel. Measure and record the insertion loss between ports 1 and 2 as P_i.

(c) The test jumper assemblies are cut in half to provide the necessary DUT pigtails required for the return loss measurements. Joint the DUT pigtail to the fiber from port 3 with a temporary joint (fusion). The DUT is then coupled to a second pigtail using an adapter. The end of the second pigtail is terminated with index gel. With the light source at the specified wavelength, measure and record the power reflection of the DUT as P'_i. Repeat steps A through C for the remaining DUT's to be measured.

(d) The return loss of each DUT is then calculated by the formula below:

$$R_L = -10\log_{10} (P'_i - P_i)/P_o + 10\log_{10} (P_2/P_3)$$

The DUT should have (as an objective) a return loss greater than 30 dB.

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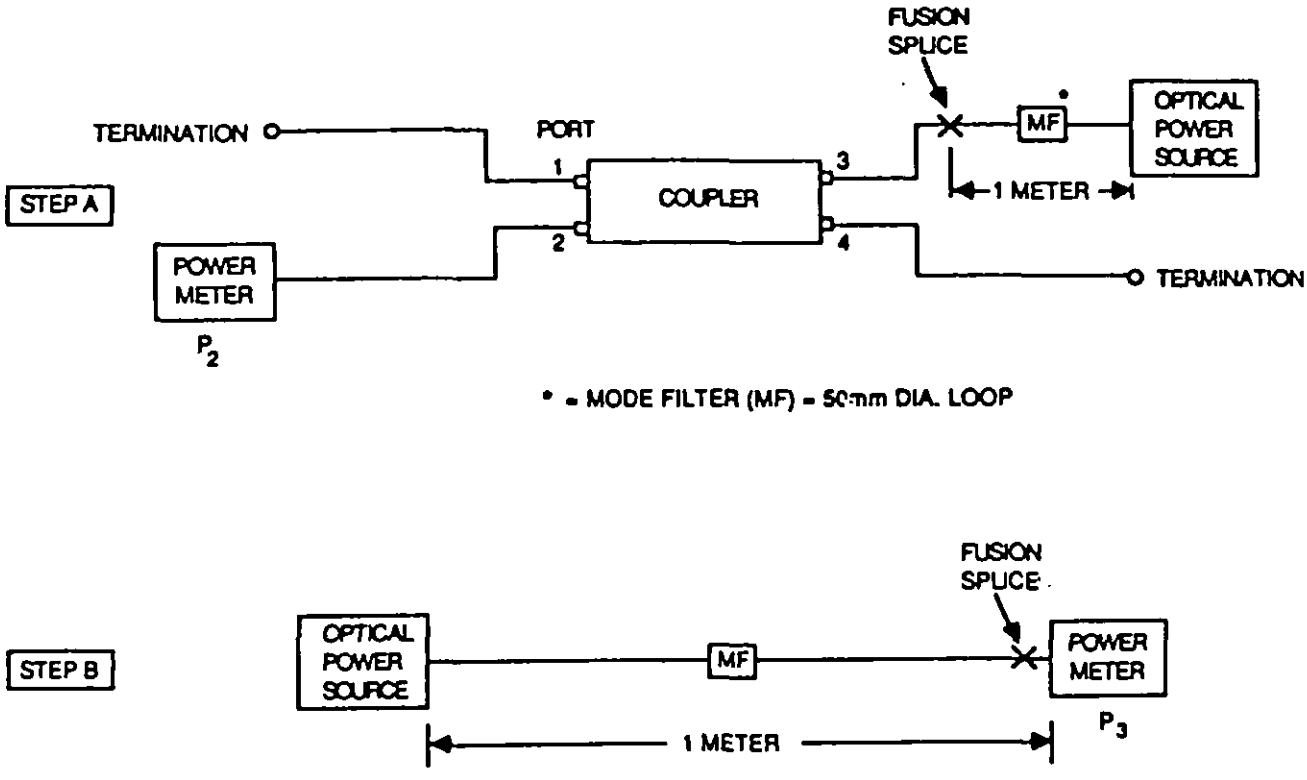


FIGURE 1. Coupler insertion loss test setup.

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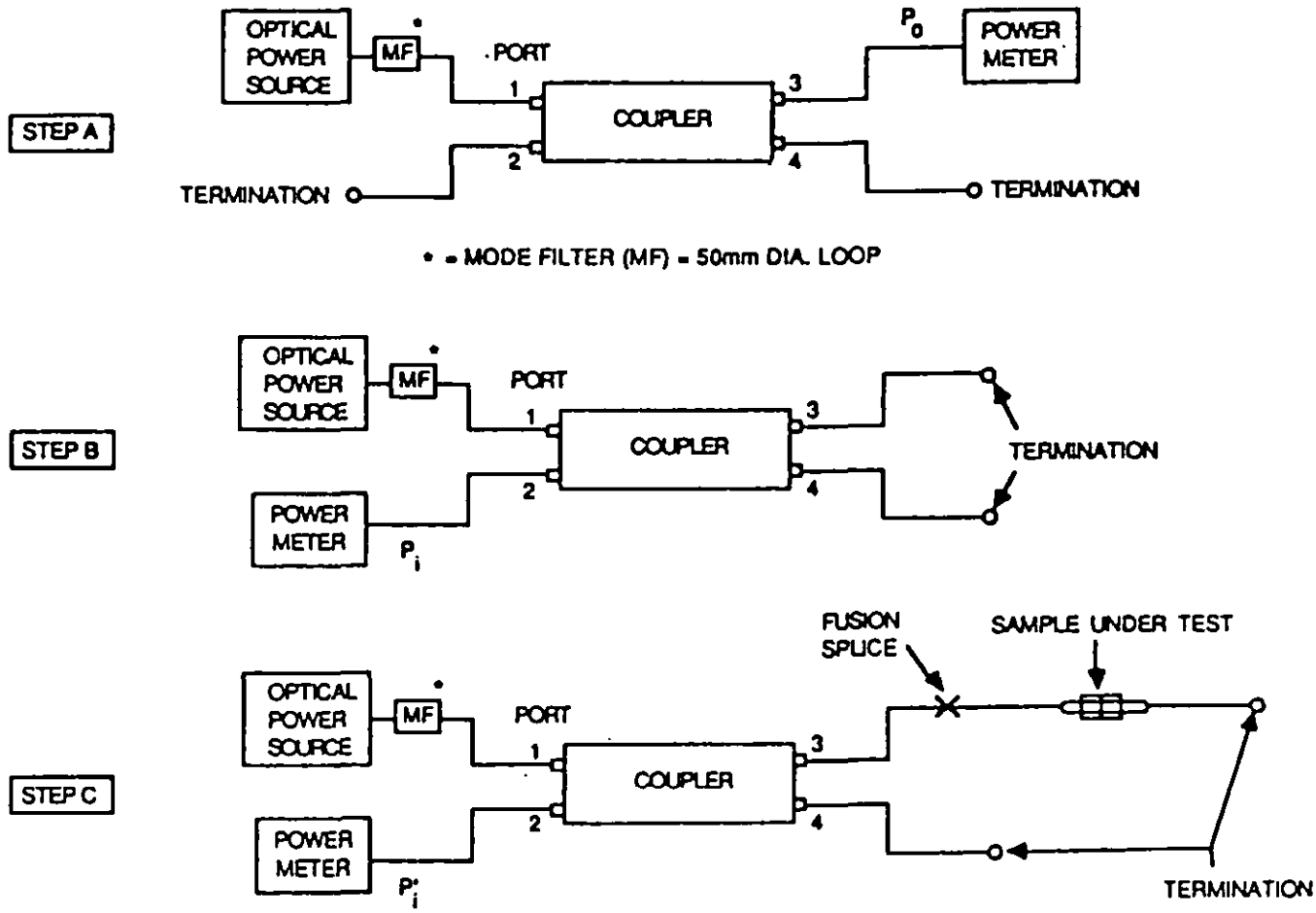


FIGURE 2. DUT return loss test setup.

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APPENDIX B

CERTIFICATION/DATA REPORT TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix covers the technical content requirements that shall be included in certification/data reports when required by the contract or order. This appendix is mandatory only when data item description DI-MISC-80678 is cited on the DD Form 1423.

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. CERTIFICATION/DATA REPORTS

30.1 Certification/data report contents. When required by the contract or order, certification/data reports shall contain the following information:

(a) Title of the test.
(b) Test sample description.
(c) Test equipment used and calibration data.
(d) Launch conditions.
(e) Test values and results.
(f) Specification sheet reference.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-24727(NAVY)	ROTARY JOINTS, FIBER OPTIC, SHIPBOARD (METRIC), GENERAL SPECIFICATION FOR

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*:
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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